

# Dinesh Kumar Rajakumaran

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🌐 LinkedIn — 🐙 GitHub — 🌐 Portfolio

📍 Wels, Austria



## Profile

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Automotive Mechatronics Engineer with experience in vehicle system simulation, control engineering, sensor evaluation, and mechatronic systems validation. Strong technical foundation in MATLAB/Simulink, model-based engineering, drivetrain control systems, vehicle dynamics, and data-driven engineering. Experienced in developing system models, executing simulation studies, analyzing technical results, and collaborating with multidisciplinary engineering stakeholders. Academic background includes MBSE, requirements specification, functional modelling, systems integration, verification and validation, project management, and automotive e-mobility. Passionate about applying simulation and structured engineering methods to develop reliable and innovative drivetrain systems.

## Skills

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- **Simulation & Model-Based Engineering:** MATLAB/Simulink, System Modelling, Vehicle Modelling, State-Space Modelling, Control System Design, Model-Based Engineering.
- **Drivetrain & Control Systems:** Powertrain Modelling, Transmission Systems, Transmission Control, Vehicle Dynamics, ABS/ESP Systems, BEV and Hybrid Control Strategies.
- **Data Analysis & Engineering Tools:** MATLAB, Python, CANoe, ROS Toolbox, Foxglove Studio, Data Processing and Visualization, MS Office, Microsoft Visio.
- **Systems Engineering:** Requirements Specification, Functional Modelling, Systems Integration, Verification & Validation, Simulation-Based Testing, Technical Documentation.
- **Automotive Systems:** Automotive Electronics, CAN Communication, Automotive Ethernet, AUTOSAR Fundamentals, Functional Safety (ISO 26262), E/E Systems.
- **Product Development:** New Product Development, Innovation Management, Project Management, Process Improvement, Engineering Analysis.
- **Professional Skills:** Stakeholder Coordination, Technical Reporting, Problem Solving, Data Analysis, Structured Decision Making, Cross-functional Communication.

## Professional Experience

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**BRP-Rotax GmbH & Co KG, Austria**

**Oct 2024 – Jun 2025**

*FH Praktikant – Control Systems Engineering*

- Developed and validated nonlinear vehicle system models for a hydrofoil-based recreational vehicle using MATLAB/Simulink.
- Planned and executed simulation studies to evaluate system behaviour and control performance under varying operating conditions.
- Analyzed simulation results, investigated system behaviour, and supported optimization of control strategies within defined project objectives.
- Coordinated technical activities with supervisors and engineering stakeholders while documenting simulation results and development.

- Developed MATLAB-based workflows for automotive sensor data evaluation using ROS Toolbox and Foxglove Studio.
- Processed and analyzed recorded datasets to evaluate sensor behaviour, identify technical issues, and support system verification.
- Configured communication and data-processing workflows for reliable acquisition, analysis, and visualization of automotive sensor data.
- Prepared technical documentation and communicated analysis results with engineering stakeholders to support project activities.

**Royal Enfield Global Headquarters, India****Mar 2022 – Sept 2023***Vehicle Testing Engineer*

- Executed functional and performance validation of ABS and braking systems on prototype and production vehicles.
- Planned vehicle testing activities, analyzed measurement data, and evaluated system performance against defined test objectives.
- Collaborated with Tier-1 suppliers including Bosch and Continental to investigate technical issues and verify corrective actions.
- Prepared technical reports, maintained validation documentation, and supported structured engineering decisions throughout product development.

**Education**

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**University of Applied Sciences Upper Austria****2023 – 2025**

M.Sc. Automotive Mechatronics and Management

*Relevant Coursework: Model-Based Engineering, Drive Train Control Systems, Drive Systems & E-Mobility, Mechatronic Systems Validation, Sensors & Micro Actuators, Automotive IT Systems, MATLAB/Simulink, Functional Safety (ISO 26262), Project Management, New Product Development, Vehicle Dynamics, Sensor Fusion, Driving Assistance Systems***Rajalakshmi Engineering College****2017 – 2021**

B.E. Automobile Engineering

*Relevant Coursework: Vehicle Dynamics, Automotive Electronics, Automotive Transmission Systems, Internal Combustion Engines, Automotive Chassis Systems, Automotive Electrical and Electronic Systems, Control Engineering, Thermodynamics, Fluid Mechanics, Finite Element Analysis (FEA), Materials Engineering, Production Technology***Projects**

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**Hydrofoil Vehicle Control and Validation (Master's Thesis)**

Developed nonlinear dynamic models of a hydrofoil vehicle including coupled vehicle dynamics and actuator behaviour. Designed and evaluated cascaded PID and Linear Model Predictive Control strategies using MATLAB/Simulink, including state-space modelling, stability analysis, actuator constraints, and performance evaluation under varying operating conditions.

**Sensor Data Evaluation using ROS and MATLAB**

Developed a MATLAB ROS Toolbox workflow for sensor data processing, visualization, and system evaluation using Foxglove Studio. Processed radar datasets and analyzed sensor behaviour to support simulation-based verification, debugging, and performance assessment of automotive sensing systems.

**Languages**

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Tamil (Native) - Fluent, English (C1) - Fluent , German (B1) - Conversational